Solution of Linear System

Augmented Matrix

An **augmented matrix** is a compact way to represent a system of **linear equations** by writing the **coefficients and constants** in a single matrix.

$$\begin{cases} 2x + 3y - z = 5 \\ x - 4y + 5z = 6 \\ 3x - y + 2z = 9 \end{cases}$$

The Augmented Matrix is

$$A_b = \begin{bmatrix} 2 & 3 & -1 & : & 5 \\ 1 & -4 & 5 & : & 6 \\ 3 & -1 & 2 & : & 9 \end{bmatrix}$$

Row Echelon Form (REF)

3 Rules

- First non-zero element in each row, called the **leading entry**, is 1
- Each leading entry is in a column to the right of the leading entry in the previous row
- Rows with all zero elements, if any, are below rows having a non-zero element

Which of the following matrices are in row echelon form?

A $\begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -6 \\ 0 & 0 & 0 \end{bmatrix}$ B $\begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ C $\xcancel{\begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$ D $\begin{bmatrix} 1 & -4 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Reduced Row Echelon Form (RREF) Rules

1️⃣ **Satisfies all REF rules**.

2️⃣ Each **leading 1** is the **only nonzero element** in its column.

✅ **Example of Reduced Row Echelon Form (RREF)**

$$\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & 4 \end{bmatrix} \qquad \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & 3 & 0 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Row Echelon and Reduced Echelon Form

The following matrices are in reduced row echelon form.

$$\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

The following matrices are in row echelon form but not reduced row echelon form.

$$\begin{bmatrix} 1 & 4 & -3 & 7 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix}, \quad \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

More on Row Echelon and Reduced Echelon Form

With any real numbers substituted for the $*$'s

Row Echelon Form

$$\begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 1 & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

Row Reduced Echelon Form

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

Exercises: determine whether the matrix is in row echelon form, reduced row echelon form, both, or neither.

a. $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

b. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix}$

c. $\begin{bmatrix} 1 & 3 & 4 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

d. $\begin{bmatrix} 1 & 5 & -3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

e. $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

f. $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 0 & 7 & 1 & 3 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

g. $\begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{bmatrix}$

Solution: Gauss Elimination Method – Working Rule

The **Gauss Elimination Method** is a step-by-step process used to **SOLVE** a system of linear equations by transforming the augmented matrix into **row echelon form** and then using **back-substitution** to find the solution

Method

Step 1: Augmented Matrix

Step 2: Convert to Row Echelon Form

Step 3: Back-Substitution

💡 **Key Notes**

✅ **Row operations allowed:**

- Swap two rows.
- Multiply a row by a nonzero constant.
- Add/subtract a multiple of one row to another.

Linear System

Three Unknown

Question No. 1 **Solve**

$$x + y + 2z = 9$$

$$2x + 4y - 3z = 1$$

$$3x + 6y - 5z = 0$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 1 \\ 0 \end{bmatrix}$$

The Augmented Matrix is

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

Gauss Elimination Method

As

$$x + y + 2z = 9$$
$$2x + 4y - 3z = 1$$
$$3x + 6y - 5z = 0$$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

$\boxed{R_2 - 2 * R_1}$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

$\boxed{R_3 - 3 * R_1}$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{bmatrix}$$

$\boxed{\frac{1}{2} * R_2}$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 3 & -11 & -27 \end{bmatrix}$$

$\boxed{R_3 - 3 * R_2}$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & -1/2 & -3/2 \end{bmatrix}$$

$\boxed{-2 * R_3}$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Echelon Form

So

$$\begin{cases} x + y + 2z = 9 \\ y - \dfrac{7}{2} z = -\dfrac{17}{2} \\ z = 3 \end{cases}$$

After substituting, we have

$$\boxed{y - \frac{7}{2}(3) = -\frac{17}{2}} \qquad \boxed{y = 2}$$

$$\boxed{x + 2 + 2(3) = 9} \qquad \boxed{x = 1}$$

$$\boxed{x = 1}$$
$$\boxed{y = 2}$$
$$\boxed{z = 3}$$

Solution: Gauss-Jordan Elimination Method – Working Rule

The **Gauss-Jordan method** is an improved version of **Gauss elimination**, used to solve a system of linear equations by transforming the augmented matrix into **reduced row echelon form (RREF)**.

Method

Step 1: Augmented Matrix

Step 2: Convert to Reduced Row Echelon Form

Step 3: Read the solution

Gauss Jordan Method

As

$$x + y + 2z = 9$$
$$2x + 4y - 3z = 1$$
$$3x + 6y - 5z = 0$$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

$\boxed{R_2 - 2 * R_1}$
$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

$\boxed{R_3 - 3 * R_1}$
$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{bmatrix}$$

$\boxed{\frac{1}{2} * R_2}$
$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 3 & -11 & -27 \end{bmatrix}$$

$\boxed{R_3 - 3 * R_2}$
$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & -1/2 & -3/2 \end{bmatrix}$$

$\boxed{-2 * R_3}$
$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$\boxed{R_1 - R_2}$
$$\begin{bmatrix} 1 & 0 & 11/2 & 35/2 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$\boxed{R_1 - 11/2 * R_3}$
$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$\boxed{R_2 + 7/2 * R_3}$
$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

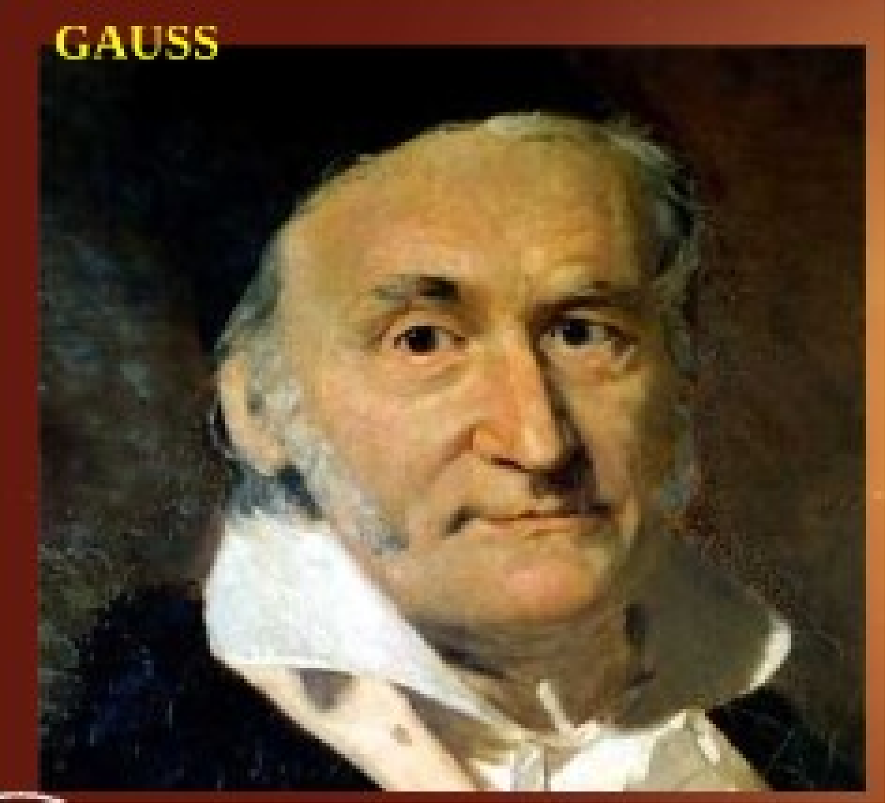
So

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

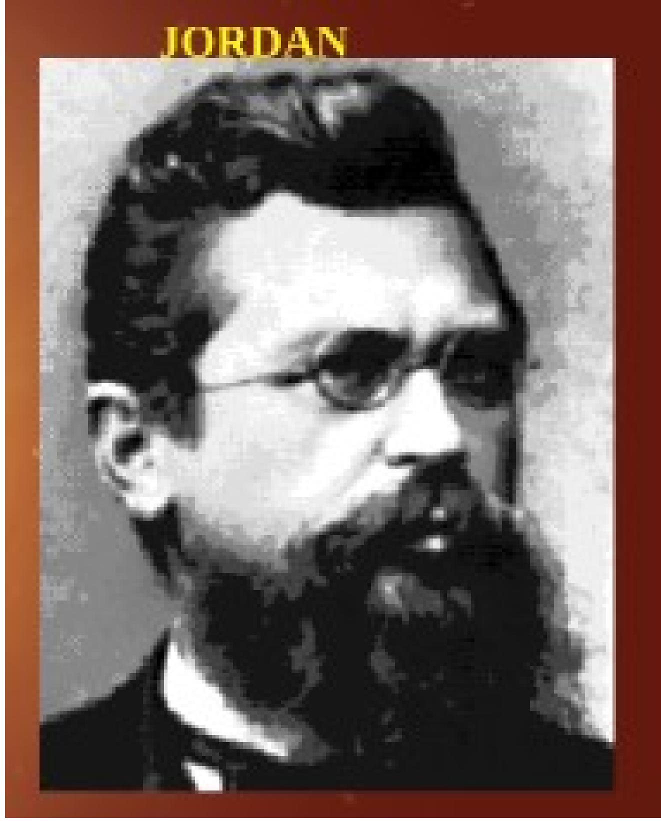
$x = 1$

$y = 2$

$z = 3$



Gauss Elimination was fully developed by German mathematician **Carl Friedrich Gauss** (1777–1855), the **Gauss-Jordan method** was a refinement made later by German Geodesist **Wilhelm Jordan** (1842–1899).



Linear System **Three variables** **With** **Infinitely Many Solutions** **Question**

$$x + 2y + 3z = 4$$
$$x + 4y + 9z = 16$$
$$2x + 4y + 6z = 8$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 9 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 16 \\ 8 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 9 \\ 2 & 4 & 6 \end{bmatrix} \qquad b = \begin{bmatrix} 4 \\ 16 \\ 8 \end{bmatrix}$$

$$Ax = b$$

$$A_b = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 9 & 16 \\ 2 & 4 & 6 & 8 \end{bmatrix} \qquad \text{RREF}(A_b) \qquad \begin{bmatrix} 1 & 0 & -3 & -8 \\ 0 & 1 & 3 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x - 3z = -8$$
$$y + 3z = 6$$
$$0z = 0$$

$$x = 3z - 8$$
$$y = -3z + 6$$
$$z = t$$

$$x = 3t - 8$$
$$y = -3t + 6$$
$$z = t$$

General solution **of the**

in **parametric equations**

Algebraic

from which all solutions can be obtained

Let $z = t$ (since it can take any real value).

So, the Final Answer (Parametric Form):

$$(3t - 8, -3t + 6, t), t \in \mathbb{R}$$

Linear System | Three variables | With | No Solution | Question

$x + y + z = 3$

$2x + 3y + z = 7$

$4x + 6y + 2z = 15$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 4 & 6 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \\ 15 \end{bmatrix}$$

$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 4 & 6 & 2 \end{bmatrix}$

$b = \begin{bmatrix} 3 \\ 7 \\ 15 \end{bmatrix}$

$Ax = b$

$$A_b = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & 3 & 1 & 7 \\ 4 & 6 & 2 & 15 \end{bmatrix}$$

$$\text{RREF}(A_b) \quad \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$x + 2z = 2$

$y - z = 1$

$0z = 1 \Rightarrow 0 = 1$

This is a **contradiction**, meaning the system is **inconsistent** and has **no solution**.

Example

Solve by Gauss–Jordan elimination.

$$\begin{aligned}
x_1 + 3x_2 - 2x_3 \phantom{{}-2x_4} + 2x_5 \phantom{{}-3x_6} &= 0\\
2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 &= -1\\
5x_3 + 10x_4 \phantom{{}+4x_5} + 15x_6 &= 5\\
2x_1 + 6x_2 \phantom{{}-5x_3} + 8x_4 + 4x_5 + 18x_6 &= 6
\end{aligned}$$

Solution The augmented matrix for the system is

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 2 & 6 & 0 & 8 & 4 & 18 & 6 \end{bmatrix}$$

Adding -2 times the first row to the second and fourth rows gives

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & -2 & 0 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 0 & 0 & 4 & 8 & 0 & 18 & 6 \end{bmatrix}$$

Multiplying the second row by -1 and then adding -5 times the new second row to the third row and -4 times the new second row to the fourth row gives

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 & 2 \end{bmatrix}$$

Interchanging the third and fourth rows and then multiplying the third row of the resulting matrix by $\frac{1}{6}$ gives the row echelon form

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Now we have to make 0 here

And here

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Adding -3 times the third row to the second row and then adding 2 times the second row of the resulting matrix to the first row yields the reduced row echelon form

$$\begin{bmatrix} 1 & 3 & 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The corresponding system of equations is

$$\begin{aligned} x_1 + 3x_2 \qquad + 4x_4 + 2x_5 \qquad &= 0 \\ x_3 + 2x_4 \qquad &= 0 \\ x_6 &= \tfrac{1}{3} \end{aligned}$$

Solving for the leading variables, we obtain

$$\begin{aligned} x_1 &= -3x_2 - 4x_4 - 2x_5 \\ x_3 &= -2x_4 \\ x_6 &= \tfrac{1}{3} \end{aligned}$$

Finally, we express the general solution of the system parametrically by assigning the free variables x_2, x_4, and x_5 arbitrary values r, s, and t, respectively. This yields

$$\boxed{x_1 = -3r - 4s - 2t}$$

$$\boxed{x_4 = s}$$

$$\boxed{x_2 = r}$$

$$\boxed{x_5 = t}$$

$$\boxed{x_3 = -2s}$$

$$\boxed{x_6 = \frac{1}{3}}$$

Exercise: Solve the systems by both Gauss and Gauss-Jordan Elimination Method

$$\boxed{x_1 + x_2 + 2x_3 = 8}$$
$$\boxed{-x_1 - 2x_2 + 3x_3 = 1}$$
$$\boxed{3x_1 - 7x_2 + 4x_3 = 10}$$

$$\boxed{2x_1 + 2x_2 + 2x_3 = 0}$$
$$\boxed{-2x_1 + 5x_2 + 3x_3 = 1}$$
$$\boxed{8x_1 + x_2 + 4x_3 = -1}$$

$$\boxed{x - y + 2z - w = -1}$$
$$\boxed{-x + 2y - 4z + w = 1}$$
$$\boxed{2x + y - 2z - 2w = -2}$$
$$\boxed{3x - 3w = -3}$$

Homogeneous Linear Systems

A system of linear equations is said to be ***homogeneous*** if the constant terms are all zero; that is, the system has the form

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + \cdots + & a_{1n}x_n & = & 0 \\ a_{21}x_1 & + & a_{22}x_2 & + \cdots + & a_{2n}x_n & = & 0 \\ \vdots & & \vdots & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + \cdots + & a_{mn}x_n & = & 0 \end{array}$$

Because a homogeneous linear system always has the trivial solution, there are only two possibilities for its solutions:

- The system has only the trivial solution.
- The system has infinitely many solutions in addition to the trivial solution.

Theorem: A homogeneous linear system with more unknowns than equations has infinitely many solutions.

In following Exercises, determine whether the homogeneous system has nontrivial solutions by inspection (without pencil and paper).

$$\begin{aligned} 2x_1 - 3x_2 + 4x_3 - x_4 &= 0 \\ 7x_1 + x_2 - 8x_3 + 9x_4 &= 0 \\ 2x_1 + 8x_2 + x_3 - x_4 &= 0 \end{aligned}$$

$$\begin{aligned} x_1 + 3x_2 - x_3 &= 0 \\ x_2 - 8x_3 &= 0 \\ 4x_3 &= 0 \end{aligned}$$

Exercise: Solve the given Homogeneous linear system by any method

$$2x_1 + x_2 + 3x_3 = 0$$
$$x_1 + 2x_2 = 0$$
$$x_2 + x_3 = 0$$

$$2x - y - 3z = 0$$
$$-x + 2y - 3z = 0$$
$$x + y + 4z = 0$$

$$v + 3w - 2x = 0$$
$$2u + v - 4w + 3x = 0$$
$$2u + 3v + 2w - x = 0$$
$$-4u - 3v + 5w - 4x = 0$$

$$x_1 + 3x_2 + x_4 = 0$$
$$x_1 + 4x_2 + 2x_3 = 0$$
$$-2x_2 - 2x_3 - x_4 = 0$$
$$2x_1 - 4x_2 + x_3 + x_4 = 0$$
$$x_1 - 2x_2 - x_3 + x_4 = 0$$

Python Code

Available on a separate word file, uploaded on Portal

How This Code Works

1. **Applies Gauss-Jordan elimination** to transform the system into row-reduced echelon form.
2. **Identifies pivot variables** (variables that have a leading 1 in a row).
3. **Identifies free variables** (columns without pivots, which lead to infinite solutions).
4. **Expresses solutions parametrically** using variables like $\mathbf{t_1, t_2, \ldots}$ when necessary

Already explained on word file with examples.


THANKS